TAHA KOULAL

Applied AI Engineer · Agentic systems & product development

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PROFILE

I build agentic AI systems and use coding agents to turn ideas into working products. Effective & efficient in how I work, I learn tools and domains quickly, starting with business needs.

A foundation in C, Unix and networking informs my system design, permissions and review of agent output.

TECHNICAL SKILLS

Applied AI & agents
Agent orchestration, MCP, Claude API, RAG, LangChain, ChromaDB, Sentence Transformers, document AI / OCR, scikit-learn, Random Forest, SHAP, prompt engineering, n8n

Backend & systems
Python, FastAPI, SQLAlchemy, PostgreSQL, REST APIs, JWT, workflow state machines, Supabase / RLS, Docker / Compose, C, C++, Unix / Linux, TCP/IP, WebSockets

Product engineering
TypeScript, JavaScript, React, React Native, Next.js, real-time systems

STRENGTHS

Business process digitisation
Scalability & reliability
Delivery with coding agents
Rapid learning & adaptation
RBAC & review gates

EDUCATION

42 Network (1337)
Software Engineering
Common Core — in progress

Validated ft_irc, Minishell, Philosophers, ft_printf, Libft and 10+ projects.

LANGUAGES

Arabic — Native
English — Professional
French — Intermediate

EXPERIENCE

PHOVA Technology · Software Engineering Intern

Phova-edge · Business operations & technical AI platform

- Analysed sales, engineering, sourcing and management workflows and handoffs; translated business needs into system design and digital workflows.
- Built authentication, user administration, opportunity management, notifications, activity history and departmental SLA tracking, including a role-gated state machine with 12 statuses and 18 transitions.
- Built the UI/UX design system and most frontend pages; implemented KPI computation. Next.js frontend with a layered FastAPI / async SQLAlchemy backend, PostgreSQL, Alembic and Docker Compose.
- Built PDF, Excel and Word extraction into structured pump data, with French / English terminology normalisation and OCR-based value localisation for review.
- Implemented RAG with LangChain, ChromaDB and Sentence Transformers; Random Forest pump classifiers with SHAP explanations, hydraulic checks and configurable language / vision providers.

UM6P — IDC Morocco · VR Development Intern

- Developed an immersive virtual visit from prototype to functional delivery using Unity and C#, including interactive 3D environments and navigation.

SELECTED PROJECTS

Factory · Multi-agent AI project factory

Claude Code · MCP · Agent orchestration

- Architected a CEO agent → specialised CTO modes → focused build agents delegation model for evaluating ideas and coordinating execution.
- Designed permission gates, milestone reviews, persistent state, audit logs, sandboxed execution and SLA-based blocker escalation.

Vamo · Football identity mobile product

React Native · Supabase · vamo-l.vercel.app

- Designed and built player identities, teams, matches, tournaments and the Football Passport, with discovery, creation, attendance and profile history flows.
- Implemented privacy-aware RLS, including teammate-only phone visibility and secure access to player data.

Transcendence · Real-time multiplayer platform

React · TypeScript · WebSockets

- Built authentication, live messaging and synchronised Pong game state; worked on frontend state and session behaviour supporting more than 20 concurrent games.

Minishell · Unix shell in C

- Built quote-aware tokenisation, parsing, variable expansion, pipelines, redirections, heredocs and builtins using fork, pipe, dup2 and execve, with interactive signal handling and descriptor cleanup.

FIT — Product concept: AI wardrobe and daily outfit recommendations using owned clothes, weather and personal preferences.
Confidential work: Two additional projects under NDA; details confidential.

Available for full-time and early-stage product opportunities